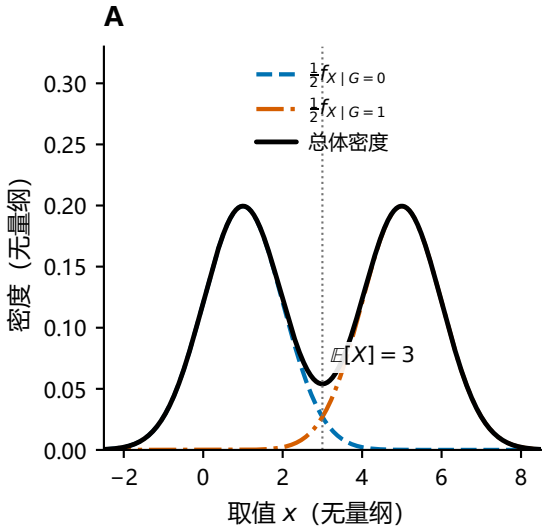
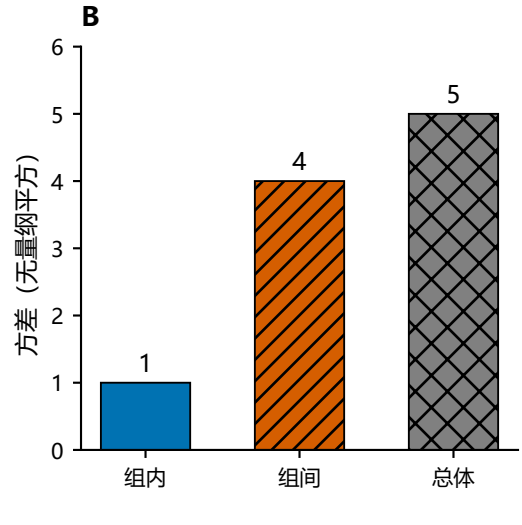


$$X \mid G=0 \sim N(1,1), \quad X \mid G=1 \sim N(5,1)$$



$$\mathrm{Var}(X)=\mathbb{E}[\mathrm{Var}(X \mid G)]+\mathrm{Var}[\mathbb{E}[X \mid G]]$$